

WorldSkills ASEAN 2025

Network Systems Administration Study Roadmap

Learning Formula

Learn the concept → Explain it yourself → Build it → Break it → Troubleshoot it

Day 1 - Networking & Routing

Topics: Default Gateway, Routing, NAT, Packet Flow, IPv4 vs IPv6. Channels: Practical Networking, Jeremy's IT Lab, NetworkChuck.

Day 2 - LDAP

Topics: Directory Services, DN, RDN, OU, UID, Groups, Authentication.

Day 2 - Certificate Authority

Topics: PKI, Root CA, CSR, Certificates, Trust Chain.

Day 3 - DNS

Topics: A, AAAA, PTR, NS, SRV, MX Records, BIND9.

Day 4 - DHCP & DDNS

Topics: DORA Process, DHCP Relay, Reservations, Scopes, DDNS, Kea DHCP.

Day 5 - Samba

Topics: SMB, Shares, Permissions, ACLs, LDAP Integration.

Day 6 - nftables

Topics: Tables, Chains, Hooks, NAT, Masquerade, Forwarding.

Day 7 - WireGuard

Topics: VPN Tunnels, Peers, Public/Private Keys, AllowedIPs.

Day 8 - Squid

Topics: Forward Proxy, Transparent Proxy, Caching, ACLs.

Day 9 - HAProxy

Topics: Frontends, Backends, Load Balancing, Health Checks, SSL Offloading.

Recommended Channels

Practical Networking, Jeremy's IT Lab, NetworkChuck.